

# Ali Darwich Ismail

## Senior Mechatronics Engineering Student

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### SUMMARY

An aspiring robotics engineer with hands-on experience in robotics design, programming, embedded systems, and control systems. Two-time international robotics competition winner and experienced robotics instructor. Seeking professional and hands-on experience to contribute to industrial robotics projects in automation, biomechatronics, control, and embedded systems.

### EDUCATION

#### Bachelor of Engineering in Mechatronics Engineering

Lebanese American University, Byblos, Lebanon

Sep 2023–Present

Expected Graduation: May 2028

#### Lebanese Baccalaureate of General Science

Rafic Hariri High School, Saida, Lebanon

Sep 2020–Jun 2023

Graduated with Distinction (16/20)

### PROFESSIONAL EXPERIENCE

#### Robotics Intern at Camerabotics, Beirut, Lebanon

February 2026 – August 2026

- Designed a UART communication system between a daisy chain of custom STM32 PCBs and a Raspberry Pi.
- Debugged and extended C++ motor control firmware on STM32, implementing position/velocity control, low-level UART protocol, and a flash saving memory map.
- Built a Python motor control API and the high-level UART protocol and deployed them on a Raspberry Pi.

#### LAU Robotics Summer Camp, Lead Organizer & Robotics Instructor

July 2025, July 2024

- Designed, delivered and monitored robotics camp sessions for 40+ students using Lego Spike Prime.
- Coordinated camp logistics and managed a team of 4 instructors and 4 volunteers.

### PROJECTS

#### FrameLock: Cinematographic robot with real-time face detection and tracking; Controller and Raspberry Pi Embedded system

##### Designer, Face detection and tracking developer

November 2025

- Applied face detection algorithm using OpenCV on a Raspberry Pi and developed the focal distance algorithm to get the distance between the robot and the face.
- Designed and tuned dual PID controllers to stabilize robot orientation and maintain smooth cinematic tracking with switchable subjects.

#### 2-DoF Biomimetic Robotic Fish; Project Leader, Embedded System Developer

May 2025

- Led the team and coordinated the project workflow.
- Designed and developed the reverse kinematics and encoder algorithms and applied them to a cascade PID motor controller.

#### Control+Farm: An Automatic farming Robot; Project Leader, MyRio and Arduino Embedded System Designer

August 2025

- Designed and programmed an autonomous robot for seeding and watering crops, using IMU data for stability monitoring and fall detection.
- Implemented automated irrigation scheduling, reducing manual labor for small-scale farming.

### TECHNICAL SKILLS

- Programming:** C++, C, Java, MATLAB, Python, Linux, ROS2, Git and GitHub.
- Robotics/Embedded Systems:** Arduino, sensors, motor control, microcontrollers, MyRio, Raspberry Pi, STM32.
- Software Tools:** AutoCAD, SolidWorks, Microsoft Office, LabView.

### AWARDS & ACHIEVEMENTS & CERTIFICATIONS

- 2x Middle East Champion, Do Your Bit Competition by BBC (2021, 2022). July 2021, July 2022
- Best Programmer, Do Your Bit National Competition (2023). July 2023
- Programming in C++: A Hands-on Introduction specialization, Coursera
- Arduino For Beginners course offered by Packt, Coursera
- Introduction to Programming with MATLAB, Coursera
- Robotics specialization offered by the University of Pennsylvania, Coursera

### LANGUAGES

- Arabic (Native)
- English (C2)
- French (B2)